SECTION 1 — Parametric Insurance Basics

Parametric insurance pays a **pre-agreed amount automatically when a measurable event (rainfall, wind speed, earthquake magnitude, etc.) crosses a predefined threshold**, instead of paying after assessing actual damage.

3 key differences from traditional insurance

Difference 1
Parametric insurance pays **based on data triggers (rainfall, wind speed)** rather than assessed loss.

Difference 2
Parametric insurance **pays automatically within days** while traditional insurance requires claims investigation.

Difference 3
Parametric insurance has **fixed payouts**, while traditional insurance reimburses **actual damages after verification**.

Real examples in India

Example 1
ICICI Lombard Parametric Weather Insurance for farmers
Source
https://www.icicilombard.com/rural-insurance/weather-insurance

Example 2
IFFCO Tokio Weather Index Insurance
Source
https://www.iffcotokio.co.in/rural-insurance/weather-insurance

Example 3
Government crop insurance using rainfall indices (PMFBY)
Source
https://pmfby.gov.in

Payout index

A payout index is the **objective measurement used to determine if the trigger event occurred**, usually calculated using meteorological data (rainfall, wind speed, temperature).
Example formula:

Rainfall Index = measured rainfall – trigger rainfall

IRDAI actively promotes **insurtech innovation**.

Examples:

Regulatory Sandbox (2019)

Allows companies to test innovative insurance products.

Source
IRDAI sandbox guidelines
https://irdai.gov.in/ADMINCMS/cms/NormalData_Layout.aspx?page=PageNo3971

This includes:

- AI underwriting
- parametric insurance
- microinsurance products

IRDAI and Microinsurance

IRDAI encourages insurance for **low-income populations**.

Microinsurance rules allow:

- small premiums
- simplified policies
- faster claims

Typical coverage limits:

Life microinsurance
₹50,000 – ₹2 lakh

Your project must comply with IRDAI rules.

Key implications:

Insurance products must be sold by licensed insurers

Your platform cannot sell insurance directly.

Instead it must partner with insurers like:

- ICICI Lombard
- HDFC ERGO
- Bajaj Allianz

Parametric insurance is allowed but regulated

IRDAI requires:

- transparent triggers
- verified external data sources
- fair payouts

Data transparency

Triggers must rely on **trusted data sources**, such as:

- IMD weather data
- satellite weather feeds
- verified APIs

Gig delivery workers in India
7.7 million
Source NITI Aayog

Average weekly income
₹4000–₹6000
Source Economic Times

Income lost due to disruptions
20–30%
Source Fairwork India

Rain days Chennai
~80
Source IMD

Insurance fraud rate
~10%
Source IRDAI

Cost of fraud
₹30,000 crore
Source Economic Times

Gig insurance market size
₹10,000 crore (estimated)
Source NITI Aayog

Claims processing cost traditional
₹1000–₹3000 per claim
Source McKinsey insurance operations

Cyclone Michaung disruption
3–4 days

India has **7.7 million gig workers today and 23.5 million expected by 2030**.

Delivery partners earn only **₹4,000–₹6,000 per week**.

Weather disruptions cause **20–30% income loss**.

Chennai experiences **~80 rainy days per year**.

Insurance fraud costs India **₹30,000 crore annually**.

Why parametric insurance fits gig workers

Gig workers lose income **due to external events like heavy rain or cyclones**, which are measurable in real time. Parametric insurance can **automatically compensate income loss without claims paperwork**, making it ideal for on-demand workers.

1. Total gig workers

77 lakh (7.7 million) platform workers in India (2020-21).
Projected 2.35 crore by 2030.

Average weekly income

₹4,000 – ₹6,000 per week

Source
Economic Times
https://economictimes.indiatimes.com/tech/startups/how-much-do-delivery-partners-earn

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3. Deliveries per day

15–20 deliveries per day

Earning per delivery

₹20–₹50 per order depending on distance and incentives

Income lost due to disruptions

20–30% income lost during heavy rain or extreme weather

Peak earning hours

Morning peak
8 AM – 11 AM

Evening peak
7 PM – 11 PM

Working days

6–7 days per week

Smartphones used

₹8,000 – ₹15,000 Android phones

IMD rainfall classification

Light
0–15.5 mm

Moderate
15.6–64.4 mm

Heavy
64.5–115.5 mm

Very Heavy
115.6–204.4 mm

Extremely Heavy

204.4 mm

Days exceeding heavy rain

~10 days per year

Cyclone Michaung

Disruption
3–4 days

Estimated income loss
₹1500–₹3000 per worker

Average AQI

Annual average AQI
~70

Worst month
January ~120

AQI above 200

Rare but occurs during severe pollution events.

Flood zones

Zone 1
Velachery

Zone 2
T Nagar

Zone 3
Pallikaranai

OpenWeatherMap API

1,000 free API calls/day

Why Chennai is ideal pilot

Chennai experiences **frequent monsoon flooding and cyclones**, making weather-triggered income loss common.

Fraud Detection

1. Common fraud types

Fake claims
Identity fraud
Policy misrepresentation

Source
Insurance Information Bureau India
https://iib.gov.in

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2. Fraud percentage

~10% of insurance claims

Source
IRDAI reports

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3. Cost of fraud

₹30,000 crore annually

Source
Economic Times
https://economictimes.indiatimes.com

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4. Location spoofing

Users fake GPS location using software to simulate presence in another area.

Source
Kaspersky security research
https://www.kaspersky.com

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5. Anomaly detection

AI models detect **behavior that deviates from normal patterns**.

Source
IBM AI fraud detection guide
https://www.ibm.com

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6. Isolation Forest

Algorithm that isolates unusual data points by randomly partitioning data.

Source
Scikit-learn documentation
https://scikit-learn.org

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7. Behavioral baseline

System learns **normal user activity patterns** and flags deviations.

Source
Deloitte fraud analytics report
https://www2.deloitte.com

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8. Zomato/Swiggy fraud signals

GPS mismatch
Unusual delivery times
Multiple account devices

Claim velocity

Number of claims filed in a short time period.

Parametric fraud detection

Techniques

Satellite verification
Weather station cross-checking
GPS validation

Parametric payouts are automatic, so **false trigger manipulation can lead to mass fraudulent payouts**.

Guidewire

1. What Guidewire does

Guidewire provides **core cloud software platforms used by insurance companies to manage policies, billing, and claims**.

Source
https://www.guidewire.com

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2. Products

PolicyCenter
Policy management

ClaimCenter
Claims processing

BillingCenter
Premium billing management

Source
Guidewire documentation

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3. Indian insurers using Guidewire

ICICI Lombard
HDFC ERGO

Revenue

$1.03 billion (FY2024)

Value proposition

End-to-end insurance core platform enabling **digital claims and policy automation**.

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6. Guidewire Marketplace

Marketplace of integrations for insurance platforms.

Source
https://marketplace.guidewire.com

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7. Gig economy involvement

Guidewire supports **usage-based and microinsurance models**.

Source
Guidewire insurtech blog

Competitors

1. Zomato insurance

Yes.

Covers

Accidental death
Hospitalization

Does NOT cover

Income loss

Swiggy insurance

Yes.

Covers

Accident insurance

Does NOT cover

Weather income loss

5. Product gaps

Gap 1
No income protection

Gap 2
No weather-trigger payouts

Gap 3
Claims processing delays

Market size

Estimated ₹10,000+ crore potential

Technical

1. OpenWeatherMap

Steps

Create account
Generate API key

Free tier

1,000 calls/day

Rainfall mm/hr

Yes

Source
https://openweathermap.org/api

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2. AQICN API

Register
Receive API token

Source
https://aqicn.org/api

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3. Flutter background location

Package

flutter_background_geolocation

Works screen off

Yes

Source
https://pub.dev

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4. Supabase

DB size
500 MB

API calls
Unlimited with rate limits

Source
https://supabase.com/pricing

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5. Render

Node hosting

Yes

Cold start

Yes (15 min idle sleep)

Reality of Gig Worker Insurance Today

The industry today mainly offers **three types of coverage**.

1. Accident insurance

Most common.

Covers:

- road accidents
- disability
- death

Example coverage
₹2–5 lakh.

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2. Health insurance

Covers hospitalization.

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3. Life insurance

Covers death benefit for family.

1. Normal Delivery Worker Earnings (Baseline)

Typical earnings:

- ~₹1000 per day for a 12–14 hour shift for some riders (reported in field interviews).

Typical order productivity assumption used in gig-economy studies:

Metric	Typical value
Deliveries/hour	3–4
Daily deliveries	18–22
Daily earnings	₹700–₹1000

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2. Impact of Rain on Traffic and Delivery Speed (Research)

Transport research shows rain significantly slows travel:

- **Traffic speed drops ~8–12% during rain**.
- **Average vehicle speed can drop by ~12 km/h during rain**.
- **Travel time increases between 8% and 140% depending on rainfall intensity**.

For delivery workers, slower traffic means **fewer orders per hour** → **lower income**.

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3. Light Rain Impact (Evidence-Based Estimate)

Light rain = **traffic capacity drop ~4–7%** and slower speeds.

Metric	Normal	Light rain
Deliveries/hour	3.5	~3.2
Daily deliveries	20	~18

Income example:

Scenario	Income
Normal	₹900
Light rain	₹800

Example effect on deliveries:Income example:**Loss: ₹100/day (~10–12%)**

Reason:

- small traffic slowdown
- slight pickup delays

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4. Moderate Rain Impact

Moderate rain causes larger delays and congestion.

Traffic research shows:

- **road capacity reduction up to ~14% under stronger rainfall**.
- travel time may increase significantly (up to 140% in worst cases).

Example delivery productivity:

Metric	Normal	Moderate rain
Deliveries/hour	3.5	~2.7
Daily deliveries	20	~15

Income example:**Loss: ₹250/day (~25–30%)**

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5. Heavy Rain / Flooded Roads (Documented Effects)

Heavy rains cause:

- road flooding
- order cancellations
- delayed deliveries
- temporary shutdown of services

Example news report:

- rain caused delivery delays and operational disruption in cities.

Typical outcome:

Metric	Heavy rain
Deliveries/day	8–12
Income	₹350–₹600

Loss: ₹400–₹600/day

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6. Additional Costs Riders Face During Rain

Delivery workers also incur extra costs:

- rain gear

- bike maintenance
- higher fuel consumption

Example report:

- riders spend about **₹300/day on fuel** from ~₹1000 earnings.

Rain increases fuel usage due to longer routes and traffic.

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7. Final Evidence-Based Loss Summary

Rain level	Deliveries/day	Income	Loss
No rain	20	₹900	—
Light rain	18	₹800	₹100
Moderate rain	15	₹650	₹250
Heavy rain	8–12	₹350–₹600	₹400–₹600